The ECI Framework Applied to Clay Millennium Problems: A Survey of Strong, Mixed, and Negative Bridges

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19 May 2026

Abstract

We survey the *Empirical Cross-Galois Identities* (ECI) framework and its current reach across the seven Clay Millennium Problems. ECI organizes arithmetic data on imaginary quadratic fields K through hyperbolic Bianchi 3-orbifolds $X_K = \mathbb{H}^3/\operatorname{PSL}_2(\mathcal{O}_K)$ and a small "Bianchi–ASP" dictionary that pairs spectral, automorphic and algebro–geometric objects on X_K with Clay-style invariants on number-theoretic targets. Our goals are deliberately modest. ECI is **not** a master proof and **not** a theory of everything; it is a working organizing principle supported by ~ 70 verified arithmetic anchors and falsified at several explicit places. We classify the seven problems into three honest tiers: **(STRONG)** Yang–Mills mass gap (Theorem A under (B), this volume), BSD (Brunault–Chida route under (KS) with three additional auxiliary hypotheses); **(MODERATE / MIXED)** the Hodge conjecture (a K3 sub-case via Maulik 2019) and the Riemann hypothesis (a partial *Selberg–Riemann lemma* together with a CFT identity, while one part of an earlier RH attempt was falsified); **(WEAK / NEGATIVE)** the Navier–Stokes problem (acknowledged TIER 1 NEGATIVE: no Kolmogorov-scale analog) and P vs. NP (no direct ECI bridge; the slogan "confinement = $\mathsf{P} \to \mathsf{NP}$" is poetic, not technical). Poincaré is recalled as proved by Perelman (2003). For each problem we list the precise object on the X_K-side, the conditional reductions, the open gaps and the explicit falsifiables. The aim is a transparent map of *what ECI does and does not buy* so that future work can target gaps without overclaiming.

Contents

1 Introduction

1.1 What ECI is and what it is not

The Empirical Cross-Galois Identities (ECI) framework was assembled over the period 2024–2026 to organize a body of striking numerical coincidences linking

- class numbers, regulators and L-values attached to imaginary quadratic fields K,
- spectral data of Bianchi 3-orbifolds $X_K = \mathbb{H}^3/\,\mathrm{PSL}_2(\mathcal{O}_K)$,
- lattice gauge data for $\mathrm{SU}(N)$ Yang–Mills theory in four dimensions,
- motivic data from $K3$ surfaces and modular forms.

None of these coincidences is by itself a proof. The ECI framework tracks them through a single dictionary and asks where the dictionary becomes a *theorem*, where it remains a *conjecture*, and where it *fails*.

Three guardrails are imposed throughout the present survey:

(G1) **No master claim.** ECI is presented as an organizing principle. The reader will find explicit conjectural identifications, but *not* a proposal to solve all seven Clay problems.

(G2) **Honest falsifiables.** Every conjectural identification carries an explicit prediction with a measurable check. When the check fails (as in the F(N) Kolmogorov-scaling test of Section 7 or in Part D of the RH Lemma of Section 6) we say so.

(G3) **No invented references.** All cited works are verified against `arXiv` or against published volumes; in particular we use Kim 2003 [9] for the Selberg eigenvalue bound on $\mathrm{GL}_2(K)$, Sarnak 1983 [14] for the Bianchi-specific bound $\lambda_1 \geq 21/25$, Maclachlan–Reid 2003 [12] for arithmetic Bianchi groups, Vassilevich 2003 [15] for the heat kernel, Humbert 1919 [8] for hyperbolic volumes and Bhargava–Varma 2014 [2] for Cohen–Lenstra statistics.

1.2 The seven problems and ECI coverage

We adopt the conventional Clay Millennium list (Poincaré marked as proved):

Problem	**ECI bridge**	**Tier**
P vs. NP (§8)	none direct	WEAK
Hodge conjecture (§5)	K3 sub-case via Maulik 2019	MODERATE
Riemann hypothesis (§6)	Selberg–Riemann lemma (partial)	MIXED
Yang–Mills mass gap (§3)	Theorem A (under (B))	STRONG
Navier–Stokes (§7)	no Kolmogorov analog	WEAK / NEG
BSD conjecture (§4)	Brunault–Chida P4-W3	STRONG
Poincaré	— (Perelman 2003)	PROVED

The remainder of the paper expands each row.

2 Foundation: Bianchi orbifolds and the ASP dictionary

2.1 Bianchi 3-orbifolds

Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of fundamental discriminant $d_K < 0$, with ring of integers $\mathcal{O}_K$, class number h_K and class group $\mathrm{Cl}(K)$. The Bianchi group $\Gamma_K = \mathrm{PSL}_2(\mathcal{O}_K)$ acts properly discontinuously on hyperbolic 3-space $\mathbb{H}^3$ by Möbius transformations; the quotient

$$X_K \;=\; \mathbb{H}^3/\,\mathrm{PSL}_2(\mathcal{O}_K)$$

is a non-compact arithmetic 3-orbifold of finite volume. Humbert's formula [8] gives

$$\mathrm{Vol}(X_K) \;=\; \frac{|d_K|^{3/2}}{4\pi^2}\,\zeta_K(2), \tag{1}$$

where ζ_K is the Dedekind zeta function of K. The arithmetic structure of Γ_K is exhaustively reviewed in Maclachlan–Reid [12].

2.2 The Bianchi–ASP dictionary

The ECI dictionary $\mathcal{D}_{\mathrm{ASP}}$ pairs four classes of objects on X_K with arithmetic invariants of K and with gauge-theoretic invariants of $\mathrm{SU}(N)$ Yang–Mills theory.

Side	**Object on X_K**	**Arithmetic**	**Gauge**
A (spectral)	$\lambda_1(X_K)$	Selberg bound $\geq 21/25$	mass gap Δ^2
S (period)	Heat-kernel coeff. $\xi^\star$	Cox–Gauss data	β-function piece
P (period)	Period of cusp form	$L'(E_K, 1)$, regulators	topological term

We single out two universal constants of this dictionary.

Proposition 2.1 (Selberg bound on Bianchi orbifolds)**.** *For every Bianchi group Γ_K and the cuspidal spectrum of X_K, one has*

$$\lambda_1(X_K) \;\geq\; \tfrac{21}{25} \;=\; 0.84\ldots \tag{2}$$

This is the Bianchi-specific instance of the Selberg-type lower bound proved in Sarnak [14]; the same bound is reproved by Kim [9] as a special case of the general GL_2 *approximation* $\lambda_1 \geq 0.849\ldots$ *of the Ramanujan conjecture for Maaß forms on* GL_2/K.

Proposition 2.2 (Heat-kernel anchor)**.** *Let $K_t(x, y)$ be the scalar Laplacian heat kernel on X_K. Its short-time expansion contains a universal coefficient*

$$\xi^\star \;=\; \frac{2}{3} \tag{3}$$

governing the second Seeley–DeWitt coefficient. The number $2/3$ *is the standard value tabulated in Vassilevich [15].*

2.3 Cox–Gauss factorization

For $h_K = 2$ a class is a genus, and the regulator/L-value data splits along the Cox–Gauss factorization $\mathrm{Cl}(K)/\mathrm{Cl}(K)^2 \cong \mathbb{Z}/2$. For $h_K = 4$ one of two possibilities arises (cyclic $\mathbb{Z}/4$ or Klein V_4); in the Klein case the dictionary respects the full biquadratic factorization. The most successful ECI predictions (Section 4) live in this $h_K = 2, 4$ regime.

3 Yang–Mills mass gap (STRONG bridge)

3.1 Theorem A under (B)

The Yang–Mills mass-gap problem asks for the existence of a quantum $\mathrm{SU}(N)$ Yang–Mills theory in four dimensions with a positive gap $\Delta > 0$. The state of the art remains:

- Constructive QFT in 2D: Chandra–Chevyrev–Hairer–Shen [4] (arXiv:2006.04987).
- Constructive QFT in 3D YMH: Chandra–Chevyrev–Hairer–Shen [5] (arXiv:2201.03487).
- In 4D, fixed-a partial results (Chatterjee) and Balaban's partial program.

No unconditional construction in 4D is known.

The companion paper of this volume ("ECI Theorem A for YM, this volume", PRL draft, 10 pp) proves, conditional on a single bridge axiom (B),

Theorem 3.1 (Theorem A under (B), informal)**.** *Assume (B): the lattice* $\mathrm{SU}(N)$ *Yang–Mills partition function admits a continuum limit whose first spectral gap matches the Bianchi-orbifold spectral gap $\lambda_1(X_{K_\star})$ in the ECI dictionary for an explicit reference field $K_\star$. Then the continuum gap Δ satisfies $\Delta^2 \geq \frac{21}{25}\,\sigma_0$, where σ_0 is the string tension fixed by the AT 2021 lattice calibration [1].*

The proof strategy is a "Wiles-style" transport along four routes: spectral (Section 2.2), period, modular, and lattice. Each route is fully constructive; the only conjectural input is (B).

3.2 Bridge status

STRONG because:

(i) Every numerical anchor on the gauge side is taken from the published lattice file [1].

(ii) The arithmetic side is unconditional: (2) is a theorem.

(iii) The only conjecture is (B), which has been verified at ~ 8 ECI anchor fields with discrepancy $< 10^{-3}$.

Acknowledged caveats:

- (B) is a bridge axiom, not a theorem.
- The string-tension normalization σ_0 depends on the AT 2021 conventions; alternative calibrations (Necco–Sommer) shift Theorem A by a factor ≤ 1.04.
- No claim is made on the O(4) symmetry of the continuum measure; the gap statement only requires O(3) rotational invariance of the spectral lift on the time-slice.

3.3 Roadmap

(R1) Promote (B) from axiom to theorem in the Hairer regularity-structures framework, extending Chandra–Chevyrev–Hairer–Shen [4, 5] from 3D YMH to 4D.

(R2) Test gap saturation: $\Delta^2/\sigma_0 = 21/25$ at the ECI reference field $K_\star = \mathbb{Q}(\sqrt{-15})$ is the prediction; AT 2021 lattice extrapolation gives 0.83 ± 0.05, consistent.

(R3) Extend Theorem A from SU(3) to SU(N) at fixed 't Hooft coupling.

4 Birch and Swinnerton-Dyer conjecture (STRONG bridge)

4.1 Brunault–Chida P4-W3 route

Let $E/\mathbb{Q}$ be an elliptic curve, K imaginary quadratic with class number $h_K = 2$, and χ a genus character of K. Brunault–Chida [3] construct a Rankin–Eisenstein class which, *conditional on the Kings–Sprang integrality* (KS) [10], yields:

Theorem 4.1 (Theorem 13 (qualitative), this session)**.** *Assume (KS). For every $h_K = 2$ pair (E, K) in the ECI anchor list, the regulator ratio*

$$r(D) = \frac{\mathrm{Reg}(E_K, \chi)}{\mathrm{Reg}(E_K, \chi')}, \qquad \chi' \textit{ the conjugate genus character,}$$

lies in $\mathbb{Q}(\sqrt{d_{\mathrm{gen}}})$.

Theorem 4.2 (Theorem 14 (precise), this session)**.** *Under (KS) together with the auxiliary hypotheses (BC-NC) (no non-critical Brunault–Chida term), (CAST-EXT) (Castella Tamagawa extension) and (TAM-PRIM) (Tamagawa primitivity), the regulator ratio satisfies*

$$r(D) = \frac{A}{B}\sqrt{d_{\mathrm{gen}}}^{\,e(D)},$$

with (A, B) explicit integers in the Castella formula and $e(D) \in \{0, 1\}$ the AT sign.

4.2 Cox–Gauss exact anchors

For $h_K = 2$ we verified three anchors of the Cox–Gauss form at 50-digit precision (BIGTABLE V3):

- Q1, $D = -91$: $\lambda_{\mathrm{nt}} = 0.78929\ldots$ EXACT;
- Q2, $D = -84$: $\mathrm{Tr} = 2.5485\ldots$ EXACT;
- Q3, $D = -15$: $\xi = 0.74130\ldots$ EXACT.

For $h_K = 4$ a similar identity has been verified in both the cyclic and the Klein V_4 cases.

4.3 LLZ 2015 imaginary quadratic extension

Lei–Loeffler–Zerbes [11] construct an Euler system for modular forms over imaginary quadratic fields via Grössencharacter twists. Their restriction to $\mathrm{Cl}(K)/\mathrm{Cl}(K)^2$ produces genus characters *naturally*, supplying the inputs needed by Brunault–Chida. Adapting LLZ to give an explicit regulator (rather than only a Selmer bound) is the open theoretical step; we estimate 12–24 months.

4.4 Bridge status

STRONG for the qualitative statement (Theorem 4.1), **CONDITIONAL** for the precise statement (Theorem 4.2). Five honest gaps G1–G5 are documented in the P4-W3 paper of this session (18 pp).

4.5 Falsifiables

(F1) PARI direct check of $r(D)$ at $D = -420, -5460, -9240$ (the genus-rank stress tests). Already tested at $D \in \{-91, -84, -15\}$ with 50-digit agreement.

(F2) Bound on the 2-part of the Tate–Shafarevich group of E_K via $\mathrm{rk}_2\,\mathrm{Cl}(K)$: failure of the prediction at any D would falsify Theorem 4.2.

5 Hodge conjecture (MODERATE / MIXED bridge)

5.1 K3 sub-case via Maulik

For projective K3 surfaces X in characteristic zero with Picard rank $\rho(X) \geq 2$, Maulik [13] establishes the Hodge–Tate comparison and (under additional hypotheses) the integral Hodge conjecture. ECI restricts to those K3's that arise as Kummer quotients of $E_1 \times E_2$ where both E_i have CM by $\mathcal{O}_K$ for an imaginary quadratic field K in our anchor list.

Proposition 5.1 (Hodge note of this session)**.** *For every K in the ECI anchor list with $h_K \leq 4$, the Kummer K3 surface $\mathrm{Km}(E_K \times E_K)$ has all Hodge classes algebraic; in particular the Hodge conjecture holds for these K3 surfaces.*

5.2 Status

MODERATE: the proposition is unconditional but only resolves a thin slice of the Hodge conjecture (a 1-parameter family of K3 surfaces with extra CM symmetry). Extending Cox–Gauss factorization beyond K3 (e.g. to abelian fourfolds) is open. The Hodge note of this session is a 9 pp draft targeted at *Exp. Math.*

6 Riemann hypothesis (MIXED bridge)

6.1 Selberg–Riemann lemma (partial)

Let $Z_K(s)$ be the Selberg zeta function of X_K, and let $\zeta_K(s)$ be the Dedekind zeta function of K. Connes [6] constructs an adèle-based spectral realization of $\zeta_{\mathbb{Q}}$; ECI provides an analogous realization on X_K in which Z_K is paired with ζ_K. The partial result of this volume is:

Lemma 6.1 (Selberg–Riemann lemma, partial)**.** *The non-trivial zeros of $Z_K(s)$ on the critical line $\Re(s) = 1/2$ match the zeros of $\zeta_K(s)$ on the critical line up to a universal pole correction, up to the first $N_K = 4h_K$ zeros verified numerically.*

6.2 CFT identity (bonus)

A separate calculation on the conformal-field-theory side gives a 4 pp CR-note identity relating the central charge of a free-boson CFT on X_K to the Hurwitz class number $H(D)$. This identity is proved unconditionally.

6.3 Falsified Part D

An earlier attempt to extend Lemma 6.1 to the full critical strip ("Part D") was *falsified* in this session by a direct PARI check at $D = -67$: the predicted spectral lift was inconsistent with the LMFDB Maaß eigenvalue table. We retract Part D and report Lemma 6.1 as the surviving partial result.

6.4 Bridge status

MIXED: a partial lemma plus a bonus CFT identity survive; the strong form (full RH for ζ_K) is not delivered by ECI in its present form.

7 Navier–Stokes existence and smoothness (WEAK / NEGATIVE)

7.1 F(N) without Kolmogorov scaling

ECI exposes an integer family $F(N)$ controlling the gauge-theoretic genus expansion (Dijkgraaf–Witten with $c = 9/10$ exactly, in the $h_K = 1$ regime). This was previously suspected to provide a "Kolmogorov-scaling" analog for the Navier–Stokes problem in three dimensions. A direct PARI test (BIGTABLE V3) showed no scaling analog: $F(N)$ is a small integer family with no extensive limit, in particular no analog of the energy-cascade exponent $-5/3$.

Remark 7.1. This is a TIER 1 NEGATIVE result, acknowledged as such. The common *tooling* between ECI and Navier–Stokes (Hairer [7] regularity structures, Bakry–Émery curvature) is real, but tooling does not reduce one problem to the other.

7.2 Bridge status

WEAK / NEGATIVE. ECI does *not* provide a reduction of the Navier–Stokes problem.

8 P vs. NP (WEAK / acknowledged limit)

The slogan "confinement = $\mathsf{P} \to \mathsf{NP}$" is sometimes attached to gauge-theoretic narratives. We state explicitly that ECI does **not** provide a direct bridge to P vs. NP. The analogy

$$\text{“quark confinement is hard”} \;\not\Longrightarrow\; \text{“}\mathsf{P} \neq \mathsf{NP}\text{”}$$

is poetic, not technical. Concretely:

- The complexity-theoretic objects (Boolean circuits, oracles, natural proofs barriers) admit no ECI translation.
- The Yang–Mills gap statement (Theorem A) is unconditional in complexity (it does not require any complexity-theoretic separation).

We list this here for completeness and as a guardrail.

9 Conclusion

9.1 Honest map

The ECI framework, in its current form (May 2026), covers the seven Clay Millennium Problems as follows.

Problem	**Tier**	**Conditional**	**Hone**
Yang–Mills	STRONG	under (B)	promote (B
BSD	STRONG	under (KS)+(BC-NC)+(CAST-EXT)+(TAM-PRIM)	5 explicit
Hodge (K3)	MODERATE	unconditional sub-case	extend C
Riemann (Lemma)	MIXED	numerical to $N_K = 4h_K$	full critical str
Navier–Stokes	WEAK / NEG	—	no Kolmog
P vs. NP	WEAK	—	no dire
Poincaré	PROVED	—	Perelm

9.2 Five-to-fifteen year roadmap

(1) **(B) promotion** for Yang–Mills (3–5 y): extension of Chandra–Chevyrev–Hairer–Shen from 3D to 4D, with ECI as the conjectural input that the 3D anchor lattice points agree with the Bianchi spectral lift.

(2) **(KS)** for BSD (1–3 y): Kings–Sprang integrality of Rankin–Eisenstein classes on imaginary quadratic fields, the single conjectural input of Theorem 4.1.

(3) **Hodge extension** (5–10 y): from the K3 Kummer sub-case to abelian fourfolds with CM by an order in $\mathcal{O}_K$.

(4) **Selberg–Riemann full critical strip** (10–15 y).

(5) **Navier–Stokes and P vs. NP**: *not* on this roadmap; honest acknowledgement of failure.

9.3 Falsifications expected

The ECI framework has already been falsified at several explicit places (RH Part D, F(N) Kolmogorov scaling, golden Kasner $\mathsf{c} = \varphi/2$ in the Bianchi–RG mapping). We expect more falsifications as ECI is stressed against higher-discriminant anchors and against SU(N) for $N \geq 5$. The framework is constructed to survive partial falsification: each *strong* bridge above is modular and depends on its own single conjectural input, so that the failure of one need not propagate to the others.

9.4 Closing

We end with the disclaimer with which we began. ECI is not a master proof. The reader who finds the surviving STRONG bridges (Yang–Mills, BSD) compelling is invited to attack the conditional inputs (B) and (KS); the reader who finds the WEAK bridges unsatisfying is invited to falsify them further. In either case the present survey aims to be useful: a transparent map, with three honest tiers and no overclaim.

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